

Compare Results

Old File:

USB_PD_R3_2 V1.0 2023-10_Part19.pdf

18 pages (883 KB)

15/04/2024 10:52:35

versus

New File:

USB_PD_R3_2 V1.1 2024-10_Chapter 9 States.pdf

20 pages (1.14 MB)

28/10/2024 11:27:52

Total Changes

316

Text only comparison

Content

- 216 Replacements
- 57 Insertions
- 43 Deletions

Styling and Annotations

- 0 Styling
- 0 Annotations

Go to First Change (page 1)

9 States and Status Reporting

9.1 Overview

This chapter describes the Status reporting mechanisms for devices with data connections (e.g., D+/D- and or SSTx+/- and SSRx+/-). It also describes the corresponding USB state a device that supports USB PD **Shall** transition to as a result of changes to the USB PD state that the device is in.

This chapter does not define the *System Policy* or the *System Policy Manager*. That is defined in [\[UCSI\]](#). In addition, the Policies themselves are not described here; these are left to the implementers of the relevant products and systems to define.

All PD Capable USB (PDUSB) *Devices* **Shall** report themselves as self-powered devices (over USB) when plugged into a PD capable *Port* even if they are entirely powered from *VBUS*. However, there are some differences between PD and [\[USB 2.0\]](#) / [\[USB 3.2\]](#); for example, the presence of *VBUS* alone does not mean that the device (*Consumer*) moves from the *USB Attached State* to the *USB Powered State*. Similarly, the removal of *VBUS* alone does not move the device (*Consumer*) from any of the USB states to the *USB Attached State*. See [Section 9.1.2, "Mapping to USB Device States"](#) for details.

PDUSB Devices **Shall** follow the PD requirements when it comes to suspend (see [Section 6.4.1.2.1.2, "USB Suspend Supported"](#)), configured, and operational power. The PD requirements when the device is configured or operational are defined in this section (see [Table 9.4, "PD Consumer Port Descriptor"](#)).

Note: The power requirements reported in the PD *Consumer Port* descriptor of the device **Shall** override the power draw reported in the *bMaxPower* field in the configuration descriptor. A *PDUSB Device* **Shall** report zero in the *bMaxPower* field after successfully negotiating a mutually agreeable *Explicit Contract* and **Shall** disconnect and re-enumerate when it switches operation back to operating in standard [\[USB 2.0\]](#), [\[USB 3.2\]](#), [\[USB4\]](#), [\[USB Type-C 2.4\]](#) or [\[USBBC 1.2\]](#). When operating in [\[USB 2.0\]](#), [\[USB 3.2\]](#), [\[USB Type-C 2.4\]](#) or [\[USBBC 1.2\]](#) mode it **Shall** report its power draw via the *bMaxPower* field.

Each *Provider* and *Consumer* will have their own Local Policies which operate between *Port Partners*. An example of a typical PD system is shown in [Figure 9.1, "Example PD Topology"](#). This example consists of a *Provider*, *Consumer/Providers* and *Consumers* connected together in a tree topology. Between directly connected devices there is both a flow of Power and also Communication consisting of both Status and Control information.

Figure 9.1 Example PD Topology

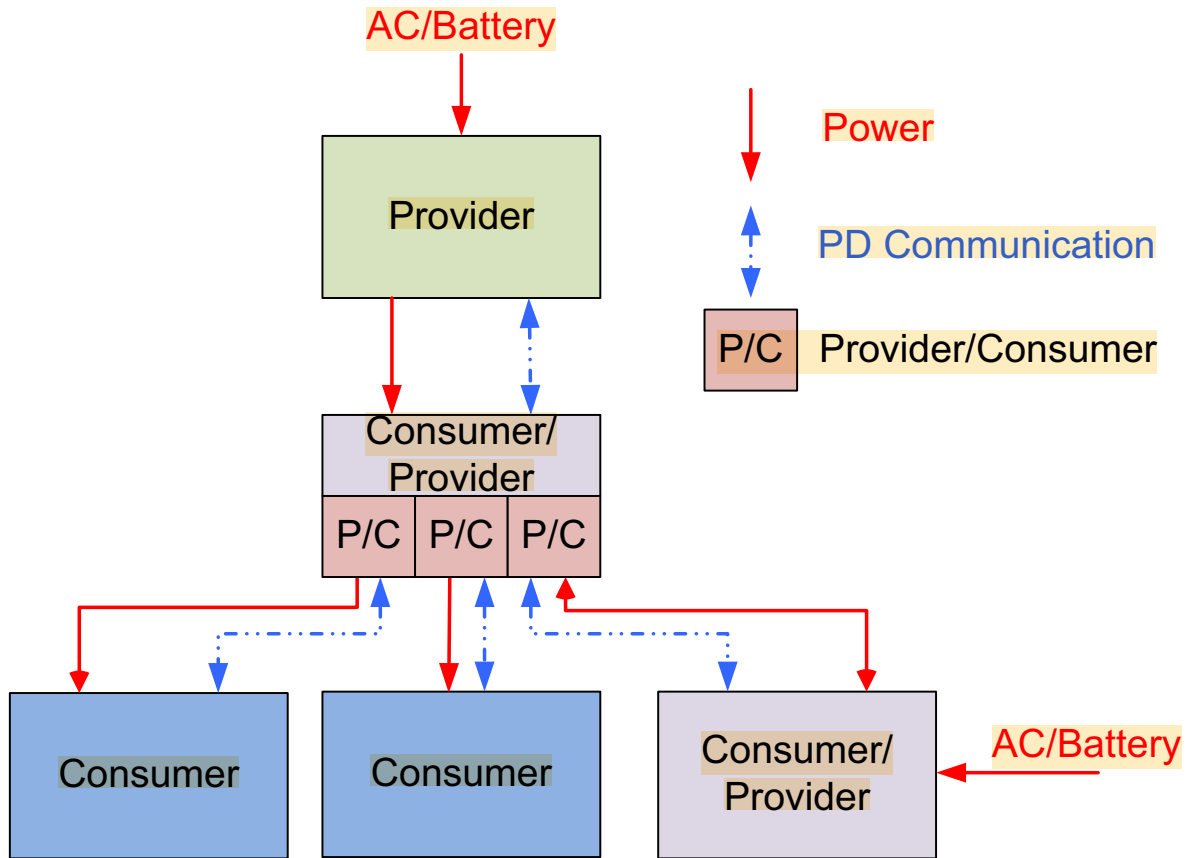
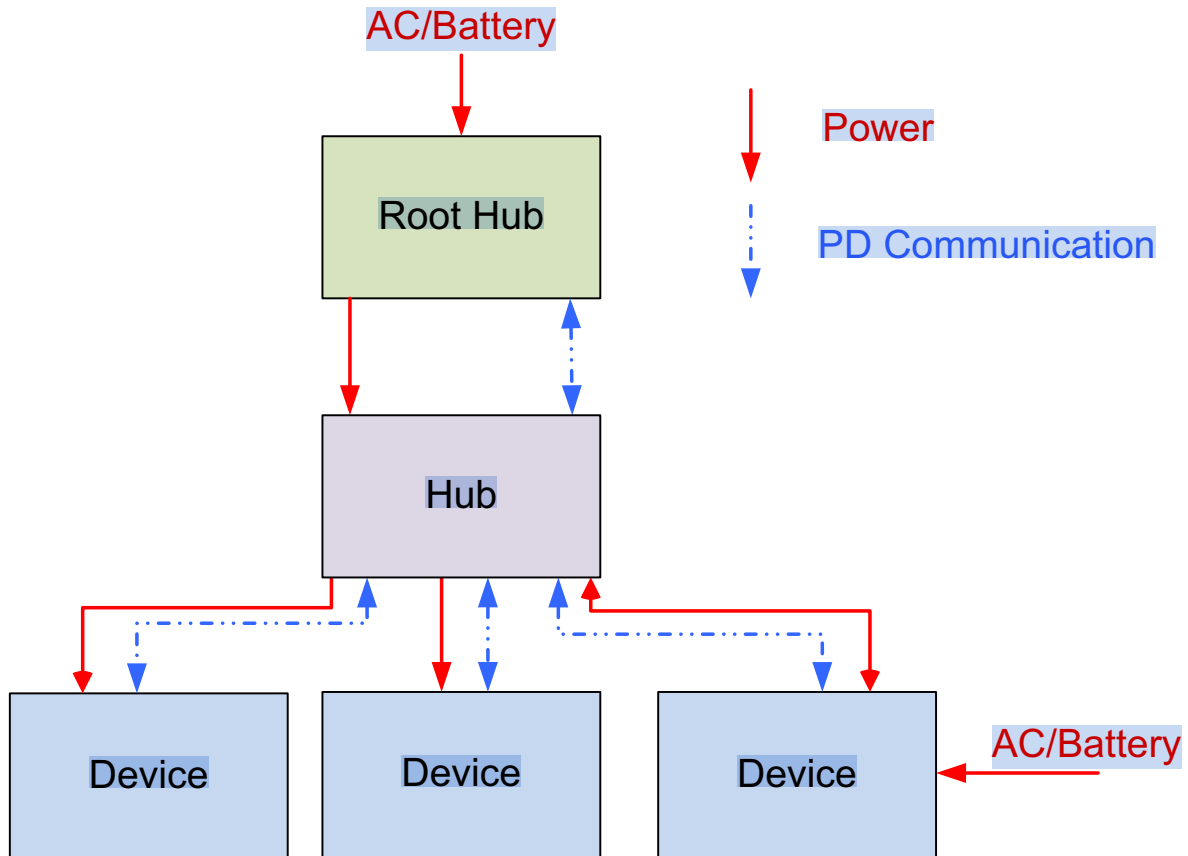


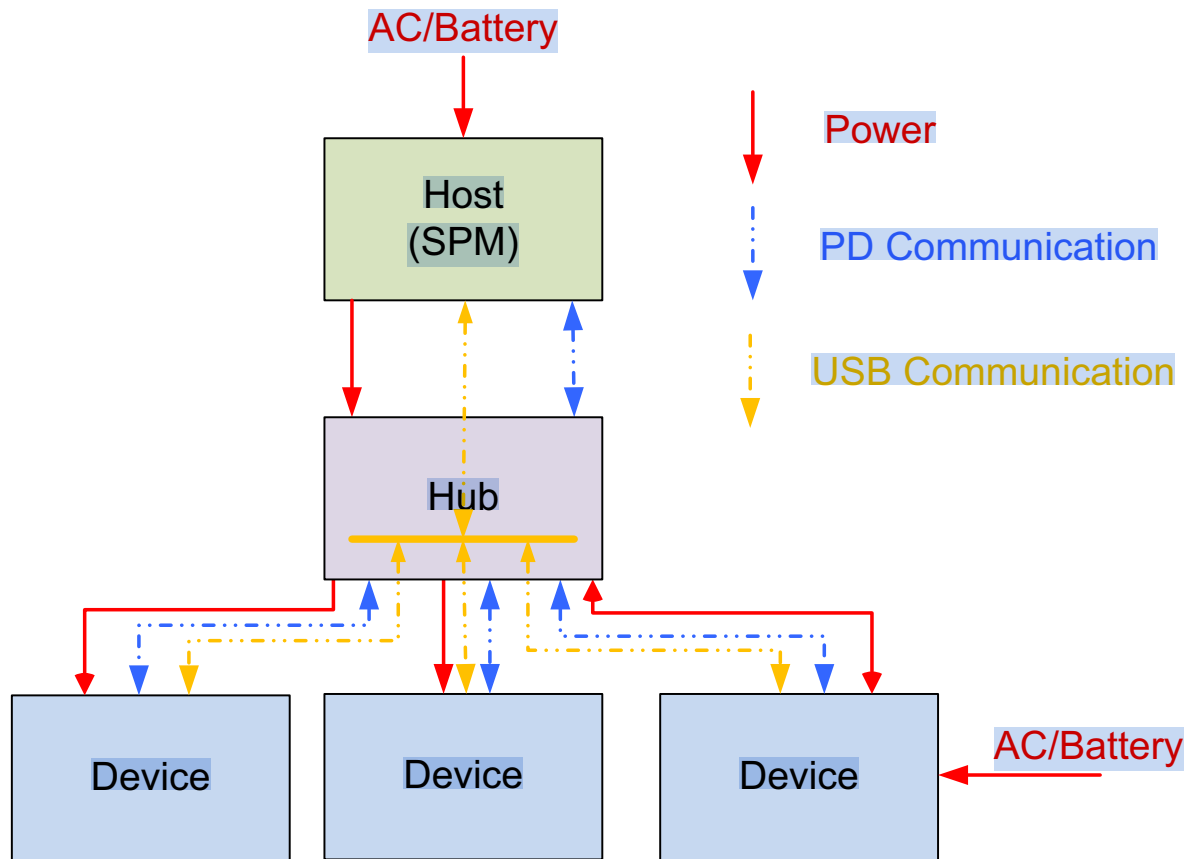
Figure 9.2, "Mapping of PD Topology to USB" shows how this same topology can be mapped to USB.

Figure 9.2 Mapping of PD Topology to USB



In a USB based system, policy is managed by the host and communication of system level policy information is via standard USB data line communication. This is a separate mechanism to the USB Power Delivery *VBUS* protocol which is used to manage *Local Policy*. When *USB Communication* is used, status information and control requests are passed directly between the *System Policy Manager (SPM)* on the host and the *Provider* or *Consumer*. See [Figure 9.3, "Use of SPM in the PD System"](#).

Figure 9.3 Use of SPM in the PD System



Status information comes from a *Provider* or *Consumer* to the *SPM* so it can better manage the resources on the host and provide feedback to the end user.

Real systems will be a mixture of devices which in terms of power management support might have implemented PD, [\[USB 2.0\]](#), [\[USB 3.2\]](#), [\[USB4\]](#), [\[USB Type-C 2.4\]](#) or [\[USBBC 1.2\]](#) or they might even just be non-compliant "power sucking devices". The level of communication of system status to the *SPM* will therefore not necessarily be comprehensive. The aim of the status mechanisms described here is to provide a mechanism whereby each connected entity in the system provides as much information as possible on the status of itself.



Information described in this section that is communicated to the *SPM* is as follows:

- Versions of *USB Type-C*[®], PD and BC supported.
- Capabilities as a *Provider/Consumer*.
- Current operational state of each *Port* e.g. Standard, *USB Type-C* Current, BC, PD and *Negotiated* power level.
- Status of AC or *Battery* Power for each *PDUSB Device* in the system.

The *SPM* can *Negotiate* with *Providers* or *Consumers* in the system in order to request a different *Local Policy*, or to request the amount of power to be delivered by the *Provider* to the *Consumer*. Any change in *Local Policy* could

trigger a *Re-negotiation* of the *Explicit Contract*, using USB Power Delivery protocols, between a directly connected *Provider* and *Consumer*. A change in how much power is available can, for example, cause a *Re-negotiation*.

9.1.1 PDUSB Device and Hub Requirements

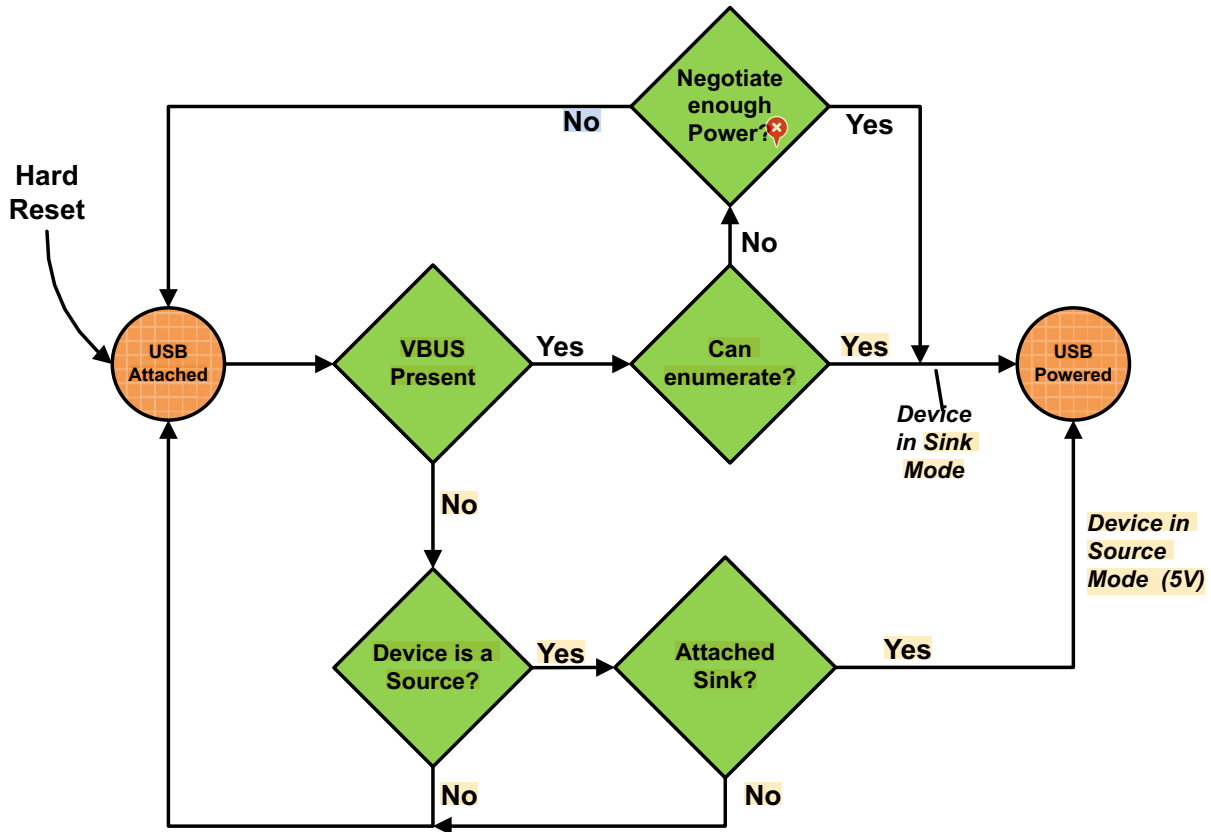
All *PDUSB Devices* **Shall** return all relevant descriptors mentioned in this chapter. *PDUSB Hubs* **Shall** also support a PD bridge as defined in *[UCSI]*.

9.1.2 Mapping to USB Device States

As mentioned in [Section 9.1, "Overview"](#) a PDUSB Device reports itself as a self-powered device. However, the device **Shall** determine whether or not it is in the USB Attached State or USB Powered States as described in [Figure 9.4, "USB Attached to USB Powered State Transition"](#), [Figure 9.5, "Any USB State to USB Attached State Transition \(When operating as a Consumer\)"](#) and [Figure 9.6, "Any USB State to USB Attached State Transition \(When operating as a Provider\)"](#). All other USB states of the PDUSB Device **Shall** be as described in Chapter 9 of [\[USB 2.0\]](#) and [\[USB 3.2\]](#).

[Figure 9.4, "USB Attached to USB Powered State Transition"](#) shows how a PDUSB Device determines when to transition from the USB Attached State to the USB Powered State. USB Type-C Dead Battery operation does not require special handling since the default state at Attach or after a Hard Reset is that the USB Device is a Sink.

Figure 9.4 USB Attached to USB Powered State Transition



[Figure 9.5, "Any USB State to USB Attached State Transition \(When operating as a Consumer\)"](#) shows how a PDUSB Device determines when to transition from the USB Powered State to the USB Attached State when the device is a Consumer. A PDUSB Device determines that it is performing a Power Role Swap as described in [Section 8.3.3.19.3, "Policy Engine in Source to Sink Power Role Swap State Diagram"](#) and [Section 8.3.3.19.4, "Policy Engine in Sink to Source Power Role Swap State Diagram"](#). See [Section 7.1.5, "Response to Hard Resets"](#) for additional information on device behavior during Hard Resets.

Figure 9.5 Any USB State to USB Attached State Transition (When operating as a Consumer)

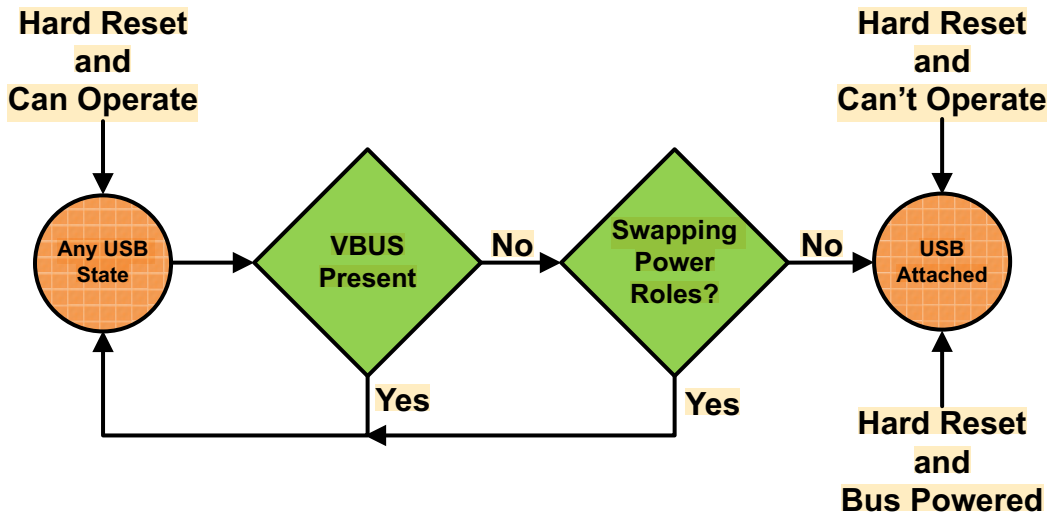


Figure 9.6, "Any USB State to USB Attached State Transition (When operating as a Provider)" shows how a PDUSB Device determines when to transition from the USB Powered State to the USB Attached State when the device is a Provider.

Figure 9.6 Any USB State to USB Attached State Transition (When operating as a Provider)

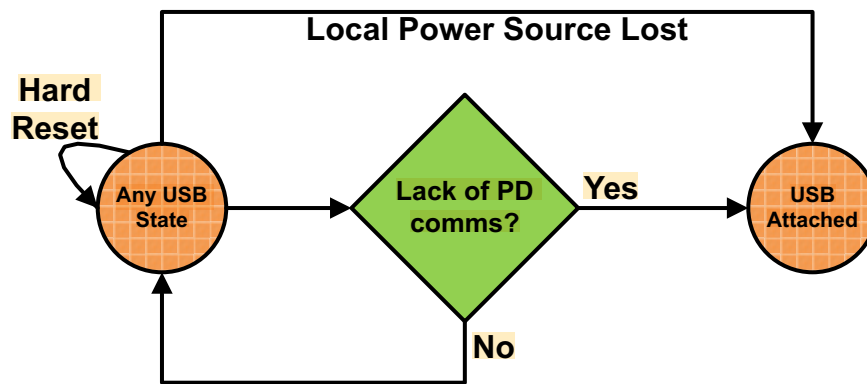
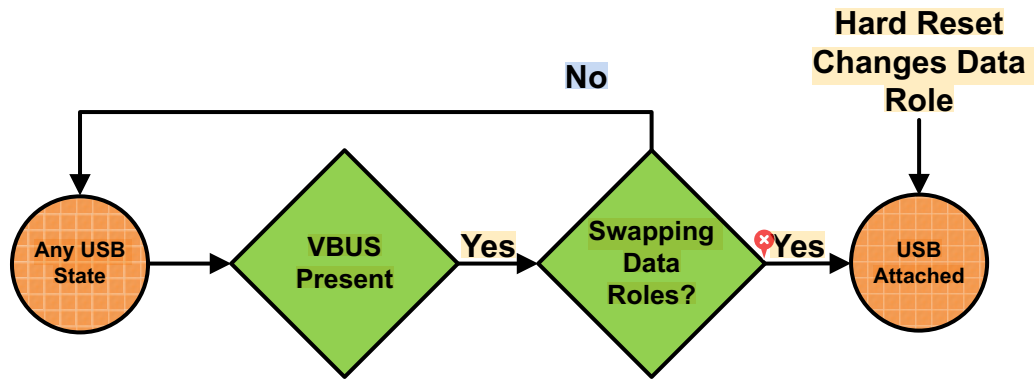


Figure 9.7, "Any USB State to USB Attached State Transition (After a USB Type-C Data Role Swap)" shows how a PDUSB Device using the USB Type-C connector determines when to transition from the USB Powered State to the USB Attached State after a Data Role Swap has been performed i.e., it has just changed from operation as a PDUSB Host to operation as a PDUSB Device. The Data Role Swap is described in [Section 6.3.9, "DR Swap Message"](#). A Hard Reset will also return a Sink acting as a PDUSB Host to PDUSB Device operation as described in [Section 6.8.3, "Hard Reset"](#). See [Section 7.1.5, "Response to Hard Resets"](#) for additional information on device behavior during Hard Resets.

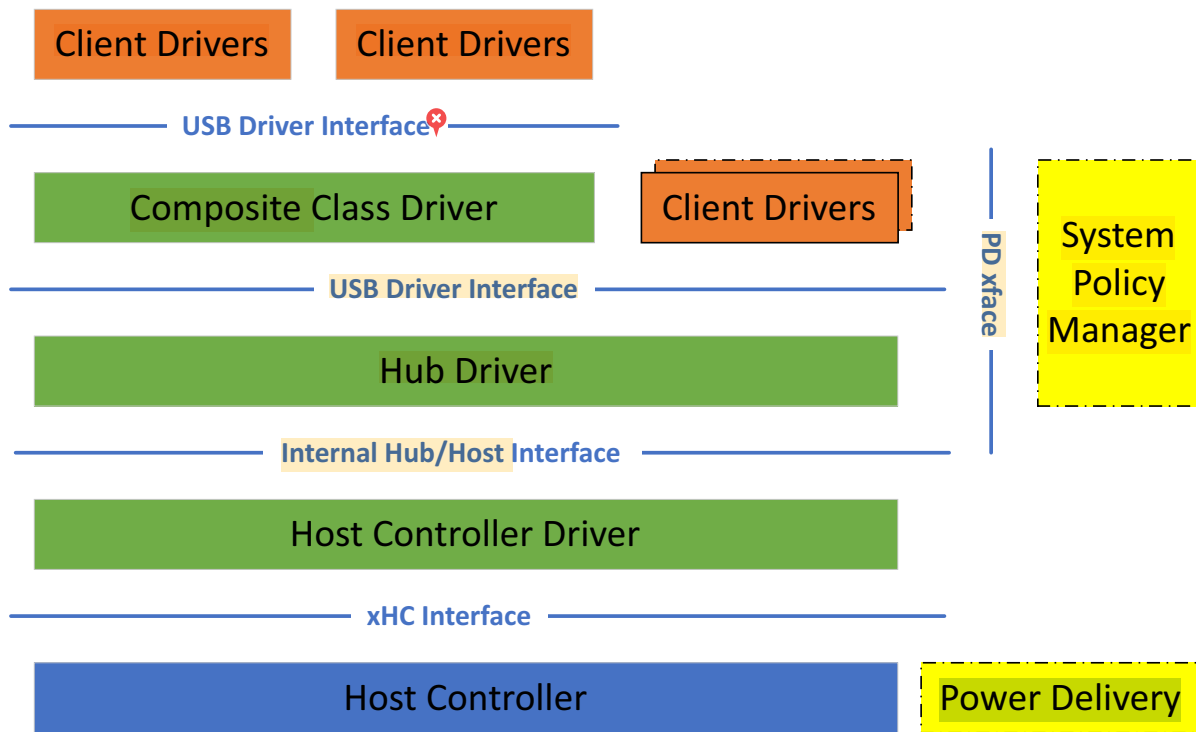
Figure 9.7 Any USB State to USB Attached State Transition (After a USB Type-C Data Role Swap)



9.1.3 PD Software Stack

[Figure 9.8, "Software stack on a PD aware OS"](#) gives an example of the software stack on a PD aware OS. In this stack we are using the example of a system with an xHCI based controller. The USB Power Delivery hardware **May** or **May Not** be a part of the xHC.

Figure 9.8 Software stack on a PD aware OS

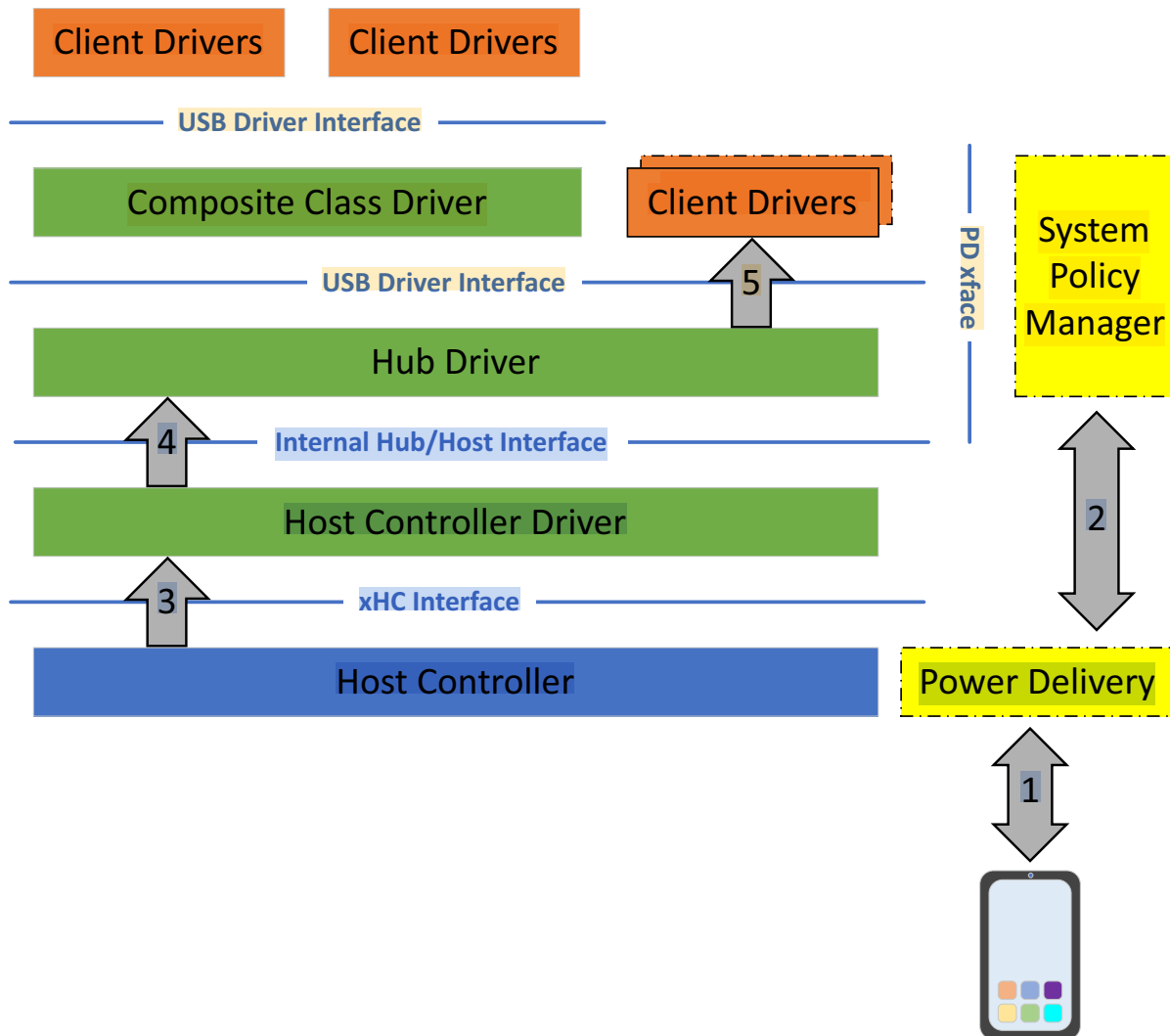


9.1.4 PDUSB Device Enumeration

As described earlier, a *PDUSB Device* acts as a self-powered device with some caveats with respect to how it transitions from the *USB Attached State* to *USB Powered State*. [Figure 9.9, "Enumeration of a PDUSB Device"](#) gives a high-level overview of the enumeration steps involved due to this change. A *PDUSB Device* will first (Step1) interact with the Power Delivery hardware and the *Local Policy* manager to determine whether or not it can get sufficient power to enumerate/operate.

*PD is likely to have established a *Explicit Contract* prior to enumeration. The *SPM* will be notified (Step 2) of the result of this *Negotiation* between the Power Delivery hardware and the *PDUSB Device*. After successfully negotiating a mutually agreeable *Explicit Contract* the device will signal a connect to the xHC. The standard USB enumeration process (Steps 3, 4 and 5) is then followed to load the appropriate driver for the function(s) that the *PDUSB Device* exposes.

Figure 9.9 Enumeration of a PDUSB Device



If a *PDUSB Device* cannot perform its intended function with the amount of power that it can get from the *Port* it is connected to, then the host system **Should** display a *notification* (on a PD aware OS) about the failure to provide sufficient power to the device. In addition, the device **Shall** follow the requirements listed in [Section 8.2.5.2.1, "Local device handling of mismatch"](#).

9.2 PD Specific Descriptors

A *PDUSB Device* **Shall** return all relevant descriptors mentioned in this section.

The device **Shall** return its capability descriptors as part of the **device's** Binary Object Store (BOS) descriptor set. [Table 9.1, "USB Power Delivery Type Codes"](#) lists the type of PD device capabilities.

Table 9.1 USB Power Delivery Type Codes

Capability Code	Value	Description
<i>POWER_DELIVERY_CAPABILITY</i>	06H	Defines the various PD Capabilities of this device
<i>BATTERY_INFO_CAPABILITY</i>	07H	Provides information on each <i>Battery</i> supported by the device
<i>PD_CONSUMER_PORT_CAPABILITY</i>	08H	The <i>Consumer</i> characteristics of a <i>Port</i> on the device
<i>PD_PROVIDER_PORT_CAPABILITY</i>	09H	The <i>Provider</i> characteristics of a <i>Port</i> on the device

9.2.1 USB Power Delivery Capability Descriptor

Table 9.2, "USB Power Delivery Capability Descriptor" details the fields in the USB **POWER_DELIVERY_CAPABILITY** Descriptor.

Table 9.2 USB Power Delivery Capability Descriptor

Offset	Field	Size	Value	Description
0	bLength	1	Number	Size of descriptor
1	bDescriptorType	1	Constant	DEVICE CAPABILITY Descriptor type
2	bDevCapabilityType	1	Constant	Capability type: POWER_DELIVERY_CAPABILITY
3	bReserved	1	Reserved	Shall be set to zero.
4	bmAttributes	4	Bitmap	Bitmap encoding of supported device level features. A value of one in a bit location indicates a feature is supported; a value of zero indicates it is not supported. Encodings are:
Bit		Description		
0		Reserved. Shall be set to zero.		
1		Battery Charging. This bit Shall be set to one to indicate this device supports USBBC 1.2 as per the value reported in the bcdBCVersion field.		
2		USB Power Delivery. This bit Shall be set to one to indicate this device supports the USB Power Delivery Specification as per the value reported in the bcdPDVersion field.		
3		Provider. This bit Shall be set to one to indicate this device is capable of providing power. This field is only Valid if Bit 2 is set to one.		
4		Consumer. This bit Shall be set to one to indicate that this device is a consumer of power. This field is only Valid if Bit 2 is set to one.		
5		This bit Shall be set to 1 to indicate that this device supports the feature CHARGING_POLICY. Note: Supporting the CHARGING_POLICY feature does not require a BC or PD mechanism to be implemented.		
6		USB Type-C Current. This bit Shall be set to one to indicate this device supports power capabilities defined in USB Type-C 2.4 as per the value reported in the bcdUSBTypeCVersion field		
7		Reserved. Shall be set to zero.		
15:8		bmPowerSource. At least one of the following bits 8, 9 and 14 Shall be set to indicate which power sources are supported.		
Bit		Description		
8		AC Supply		
9		Battery		
10		Other		
13:11	NumBatteries. This field Shall only be Valid when the Battery field is set to one and Shall be used to report the number of batteries in the device.			
14	Uses VBUS			
15	Reserved and Shall be set to zero.			
13:16	Reserved. Shall be set to zero.			

Table 9.2 USB Power Delivery Capability Descriptor

Offset	Field	Size	Value	Description
8	bcdBCVersion	2	BCD	Battery Charging Specification Release Number in Binary-Coded Decimal (e.g., V1.20 is 120H). This field shall only be Valid if the device indicates that it supports [USBBC 1.2] in the bmAttributes field.
10	bcdPDVersion	2	BCD	USB Power Delivery Specification Release Number in Binary-Coded Decimal. This field shall only be Valid if the device indicates that it supports PD in the bmAttributes field.
12	bcdUSBTypeCVersion	2	BCD	USB Type-C Specification Release Number in Binary-Coded Decimal. This field shall only be Valid if the device indicates that it supports USB Type-C in the bmAttributes field.

9.2.2 Battery Info Capability Descriptor

A PDUSB Device **shall** support the capability descriptor shown in [Table 9.3, "Battery Info Capability Descriptor"](#) if it reported that one of its power sources was a *Battery* in the bmPowerSource field in its Power Deliver Capability Descriptor. It **shall** return one **BATTERY_INFO_CAPABILITY** Descriptor per *Battery* it supports.

Table 9.3 Battery Info Capability Descriptor

Offset	Field	Size	Value	Description
0	bLength	1	Number	Size of descriptor
1	DescriptorType	1	Constant	DEVICE CAPABILITY Descriptor type
2	DevCapabilityType	1	Constant	Capability type: BATTERY_INFO_CAPABILITY
3	iBattery	1	Index	Index of string descriptor shall contain the user-friendly name for this <i>Battery</i> .
4	iSerial	1	Index	Index of string descriptor shall contain the Serial Number String for this <i>Battery</i> .
5	iManufacturer	1	Index	Index of string descriptor shall contain the name of the Manufacturer for this <i>Battery</i> .
6	bBatteryId	1	Number	Value shall be used to uniquely identify this <i>Battery</i> in status Messages.
7	Reserved	1	Number	Reserved and shall be set to zero.
8	dwChargedThreshold	4	mWh	shall contain the <i>Battery</i> charge value above which this <i>Battery</i> is considered to be fully charged but not necessarily "topped off."
12	dwWeakThreshold	4	mWh	shall contain the minimum charge level of this <i>Battery</i> such that above this threshold, a device can be assured of being able to power up successfully (see [USBBC 1.2]).
16	dwBatteryDesignCapacity	4	mWh	shall contain the design capacity of the <i>Battery</i> .
20	dwBatteryLastFullchargeCapacity	4	mWh	shall contain the maximum capacity of the <i>Battery</i> when fully charged.

9.2.3 PD Consumer Port Capability Descriptor

A PDUSB Device **Shall** support the **PD_CONSUMER_PORT_CAPABILITY** descriptor shown in [Table 9.4, "PD Consumer Port Descriptor"](#) if it is a Consumer.

Table 9.4 PD Consumer Port Descriptor

Offset	Field	Size	Value	Description
0	bLength	1	Number	Size of descriptor
1	bDescriptorType	1	Constant	DEVICE CAPABILITY Descriptor type
2	bDevCapabilityType	1	Constant	Capability type: <i>PD_CONSUMER_PORT_CAPABILITY</i>
3	bReserved	1	Number	<i>Reserved</i> and <i>Shall</i> be set to zero.
4	bmCapabilities	2	Bitmap	Capability: This field <i>Shall</i> indicate the specification the <i>Consumer Port</i> will operate under.
6	wMinVoltage	2	Number	<i>Shall</i> contain the minimum voltage in 50mV units that this <i>Consumer</i> is capable of operating at.
8	wMaxVoltage	2	Number	<i>Shall</i> contain the maximum voltage in 50mV units that this <i>Consumer</i> is capable of operating at.
10	wReserved	2	Number	<i>Reserved</i> and <i>Shall</i> be set to zero.
12	dwMaxOperatingPower	4	Number	<i>Shall</i> contain the maximum power in 10mW units this <i>Consumer</i> can draw when it is in a steady state operating mode.
16	dwMaxPeakPower	4	Number	<i>Shall</i> contain the maximum power in 10mW units this <i>Consumer</i> can draw for a short duration of time (<i>dwMaxPeakPowerTime</i>) before it falls back into a steady state.
20	dwMaxPeakPowerTime	4	Number	<i>Shall</i> contain the time in 100ms units that this <i>Consumer</i> can draw peak current. A device <i>Shall</i> set this field to 0xFFFF if this value is unknown.

9.2.4 PD Provider Port Capability Descriptor

A PDUSB Device **Shall** support the **PD_PROVIDER_PORT_CAPABILITY** descriptor shown in [Table 9.5, "PD Provider Port Descriptor"](#) if it is a Provider.

Table 9.5 PD Provider Port Descriptor

Offset	Field	Size	Value	Description
0	bLength	1	Number	Size of descriptor
1	bDescriptorType	1	Constant	DEVICE CAPABILITY Descriptor type
✖✖ 2	bDevCapabilityType	1	Constant	Capability type: <i>PD_PROVIDER_PORT_CAPABILITY</i>
✖✖ 3	bReserved	1	Number	Reserved and Shall be set to zero.
4	bmCapabilities	2	Bitmap	Capability: This field Shall indicate the specification the <i>Provider Port</i> will operate under.
Bit				Description
0				Battery Charging (BC)
1				USB Power Delivery (PD)
2				<i>USB Type-C</i> Current
	15:3 ✖	Reserved and Shall be set to zero.		
✖ 6	bNumOfPDObjects	1	Number	Shall indicate the number of <i>Power Data Objects</i> .
7	bReserved	1	Number	Reserved and Shall be set to zero.
8	wPowerDataObject1	4	Bitmap	Shall contain the first <i>Power Data Object</i> supported by this <i>Provider Port</i> . See Section 6.4.1, "Capabilities Message" for details of the <i>Power Data Objects</i> .
...	...	...	...	...
4*(N+1)	wPowerDataObjectN	4	Bitmap	Shall contain the 2nd and subsequent <i>Power Data Objects</i> supported by this <i>Provider Port</i> . See Section 6.4.1, "Capabilities Message" for details of the <i>Power Data Objects</i> .

9.3 PD Specific Requests and Events

A *PDUSB Device* that is compliant to this specification **Shall** support the *Battery* related requests if it has a *Battery*.

A *PDUSB Hub* that is compliant to this specification **Shall** support a USB PD Bridge as described in [UCSI] irrespective of whether the *PDUSB Hub* is a *Provider*, a *Consumer*, or both.

9.3.1 PD Specific Requests

PD defines requests to which *PDUSB Devices* **Shall** respond as outlined in [Table 9.6, "PD Requests"](#). All **Valid** requests in [Table 9.6, "PD Requests"](#) **Shall** be implemented by *PDUSB Devices*.

Table 9.6 PD Requests

Request	bmRequestType	bRequest	wValue	wIndex	wLength	Data
GetBatteryStatus	10000000B	GET_BATTERY_STATUS	Zero	Battery ID	Eight	Battery Status
SetPDFeature	00000000B	set_feature	Feature Selector	Feature Specific	Zero	None

[Table 9.7, "PD Request Codes"](#) gives the bRequest values for *Commands* that are not listed in the hub/device framework chapters of [USB 2.0], [USB 3.2].

Table 9.7 PD Request Codes

bRequest	Value
GET_BATTERY_STATUS	21

[Table 9.8, "PD Feature Selectors"](#) gives the **Valid** feature selectors for the PD class. Refer to [Section 9.4.2.1, "BATTERY_WAKE_MASK Feature Selector"](#), and [Section 9.4.2.2, "CHARGING_POLICY Feature Selector"](#) for a description of the features.

Table 9.8 PD Feature Selectors

Feature Selector	Recipient	Value
BATTERY_WAKE_MASK	Device	40
CHARGING_POLICY	Device	54

9.4 PDUSB Hub and PDUSB Peripheral Device Requests

9.4.1 GetBatteryStatus

The request shown in [Table 9.9, "Get Battery Status Request"](#) returns the current status of the *Battery* in a *PDUSB Hub/Peripheral*, with *Battery Status* information as shown in [Table 9.10, "Battery Status Structure"](#).

Table 9.9 Get Battery Status Request

bmRequestType	bRequest	wValue	wIndex	wLength	Data
10000000B	GET_BATTERY_STATUS	Zero	Battery ID	Eight	Battery Status

Table 9.10 Battery Status Structure

Offset	Field	Size	Value	Description	
0📌	bBatteryAttributes	1	Number	Shall indicate whether a <i>Battery</i> is installed and whether this is charging or discharging.	
				Value	Description
				0	There is no <i>Battery</i>
				1	The <i>Battery</i> is charging
				2	The <i>Battery</i> is discharging
				3	The <i>Battery</i> is neither discharging nor charging
255...4	Reserved and Shall Not be used				
1	bBatterySOC	1	Number	Shall indicate the Battery State of Charge given as percentage value from Battery Remaining Capacity.	
2📌	bBatteryStatus	1	Number	If a <i>Battery</i> is present Shall indicate the present status of the <i>Battery</i> .	
				Value	Description
				0	No error
				1	<i>Battery</i> required and not present
				2	<i>Battery</i> non-chargeable/wrong chemistry
				3	Over-temp shutdown
				4	Over-voltage shutdown
				5	Over-current shutdown
				6	Fatigued <i>Battery</i>
				7	Unspecified error
255...8	Reserved and Shall Not be used				

Table 9.10 Battery Status Structure

Offset	Field	Size	Value	Description	
3	bRemoteWakeCapStatus	1	Bitmap	If the device supports remote wake, then the device Shall support Battery Remote wake events. The default value for the Remote wake events Shall be turned off (set to zero) and can be enable/disabled by the host as required. If set to one the device Shall generate a wake event when a change of status occurs. See Section 9.4.2, "SetPDFeature" for more details.	
				Value	Description
				0	Battery present event
				1	Charging flow
				2	Battery error
7:3	Reserved and Shall be set to zero				
4	wRemainingOperatingTime	2	Number	Shall contain the operating time (in minutes) until the Weak Battery threshold is reached, based on Present Battery Strength and the device's present operational power needs. Note: This value Shall exclude any additional power received from charging. A Battery that is not capable of returning this information Shall return a value of 0xFFFF.	
6	wRemainingChargeTime	2	Number	Shall contain the remaining time (in minutes) until the Charged Battery threshold is reached based on Present Battery Strength, charging power and the device's present operational power needs. Value Shall only be Valid if the Charging Flow is "Charging". A Battery that is not capable of returning this information Shall return a value of 0xFFFF.	

If wValue or wLength are not as specified above, then the behavior of the *PDUSB Device* is not specified.

If wIndex refers to a Battery that does not exist, then the *PDUSB Device* **Shall** respond with a Request Error.

If the *PDUSB Device* is not configured, the *PDUSB Hub's* response to this request is undefined.

If the *PDUSB Hub* is not configured, the *PDUSB Hub's* response to this request is undefined.

9.4.2 SetPDFeature

The request shown in [Table 9.11, "Set PD Feature"](#) sets the value requested in the *PDUSB Hub/Peripheral*.

Table 9.11 Set PD Feature

bmRequestType	bRequest	wValue	wIndex	wLength	Data
00000000B	set_feature	Feature Selector	Feature Specific	Zero	None

Setting a feature enables that feature or starts a process associated with that feature; see [Table 9.8, "PD Feature Selectors"](#) for the feature selector definitions. Features that **May** be set with this request are:

- BATTERY_WAKE_MASK.
- CHARGING_POLICY.

9.4.2.1 BATTERY_WAKE_MASK Feature Selector

* When the feature selector is set to BATTERY_WAKE_MASK, then the wIndex field is structured as shown in [Table 9.12, "Battery Wake Mask"](#).

Table 9.12 Battery Wake Mask

Bit	Description
0	Battery Present: When this bit is set then the <i>PDUSB Device</i> Shall generate a wake event if it detects that a Battery has been inserted.
1	Charging Flow: When this bit is set then the <i>PDUSB Device</i> Shall generate a wake event if it detects that a Battery switched from charging to discharging or vice versa.
2	Battery Error: When this bit is set then the <i>PDUSB Device</i> Shall generate a wake event if the Battery has detected an error condition.
15:3	Reserved and Shall Not be used

The *SPM* **May** Enable or Disable the wake events associated with one or more of the above events by using this feature.

If the *PDUSB Hub* is not configured, the *PDUSB Hub*'s response to this request is undefined.



When the feature selector is set to CHARGING_POLICY, the wIndex field **Shall** be set to one of the values defined in [Table 9.13, "Charging Policy Encoding"](#). If the device is using *USB Type-C* Current above the default value or is using PD then this feature setting has no effect and the rules for power levels specified in the [\[USB Type-C 2.4\]](#) or USB PD specifications **Shall** apply.

Table 9.13 Charging Policy Encoding

Value	Description
00H	The device Shall follow the default current limits as defined in the USB 2.0 or USB 3.1 specification, or as negotiated through other USB mechanisms such as BC. This is the default value.
01H	The Device May draw additional power during the unconfigured and suspend states for the purposes of charging. For charging the device itself, the device Shall limit its current draw to the higher of these two values: ICCHPF as defined in the USB 2.0 or USB 3.1 specification, regardless of its USB state. Current limit as negotiated through other USB mechanisms such as BC.
02H	The Device May draw additional power during the unconfigured and suspend states for the purposes of charging. For charging the device itself, the device Shall limit its current draw to the higher of these two values: ICCLPF as defined in the USB 2.0 or USB 3.1 specification, regardless of its USB state. Current limit as negotiated through other USB mechanisms such as BC.
03H	The device Shall Not consume any current for charging the device itself regardless of its USB state.
04H-FFFFH	Reserved and Shall Not be used

This is a **Valid Command** for the *PDUSB Hub/Peripheral* in the Address or Configured USB states. Further, it is only **Valid** if the device reports a USB PD capability descriptor in its BOS descriptor and Bit 5 of the bmAttributes in that descriptor is set to 1. The device will go back to the wIndex default value of 0 whenever it is reset.